Accelerating Discovery,
Empowering Scientists

THE PROBLEM

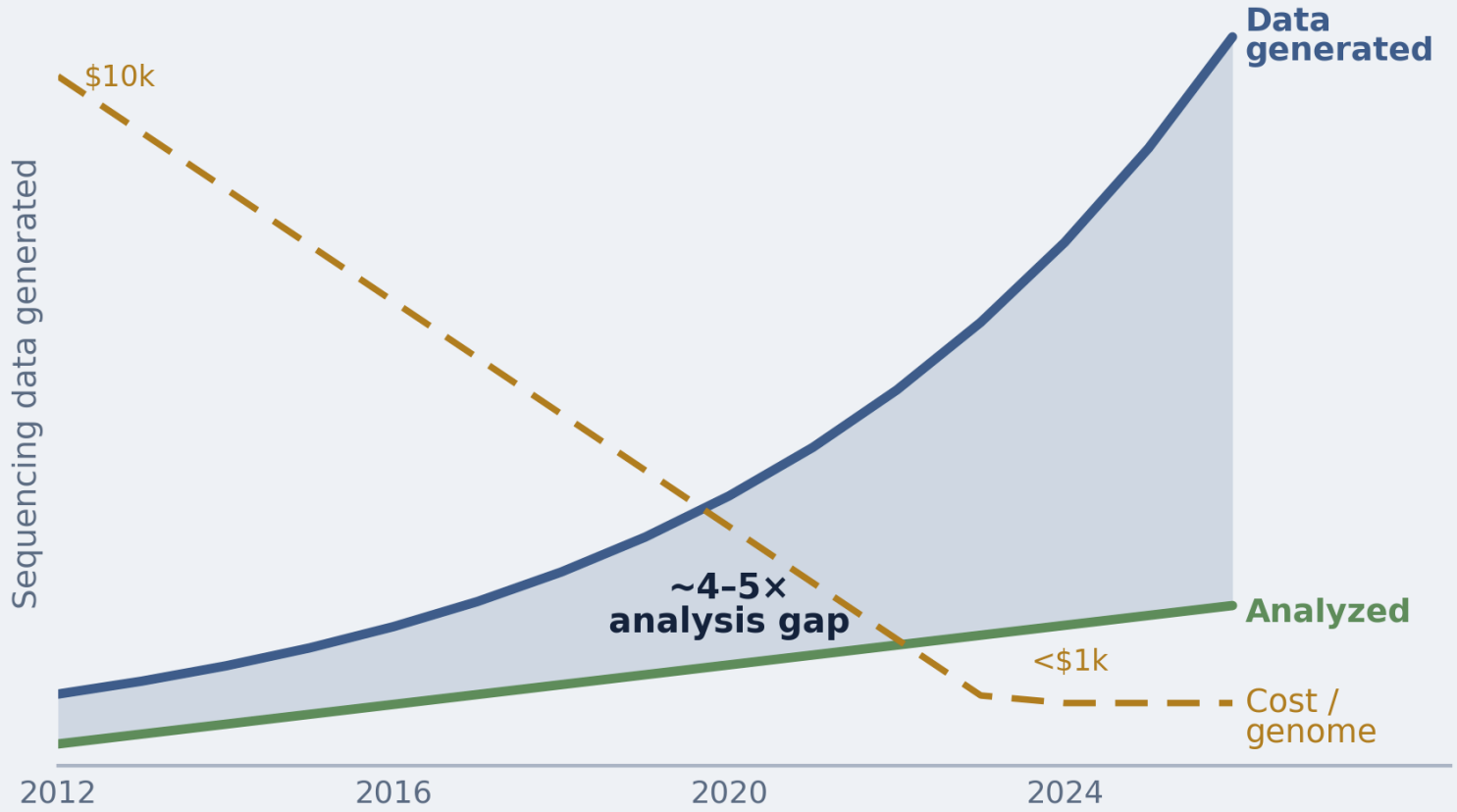
The cell is the original black box.

The average hides the cells that matter.

80%+
of labs lack the expert support to interpret it

4–5×
gap between data generated vs. analyzed

$5–20k
to outsource a single data analysis



Sources: cost/genome, NHGRI. Analysis gap illustrative.

THE SOLUTION

From raw data to answers you can defend.



Private & secure | Every step logged | Nothing can be quietly changed

Human-validated pipelines do the science. The AI orchestrates them inside a registered plan and writes the report.

HOW IT WORKS · THE PRAP

AI orchestrates. Humans validate. The plan governs.

Same data + same code + same plan → the same verdict.



Provenance proves it ran. Prairie proves it's right. The signature says a scientist stands behind it.

BUILT ON AWS

Native to the prize.

The proof of concept extends the stack that is already live. It does not replace it.

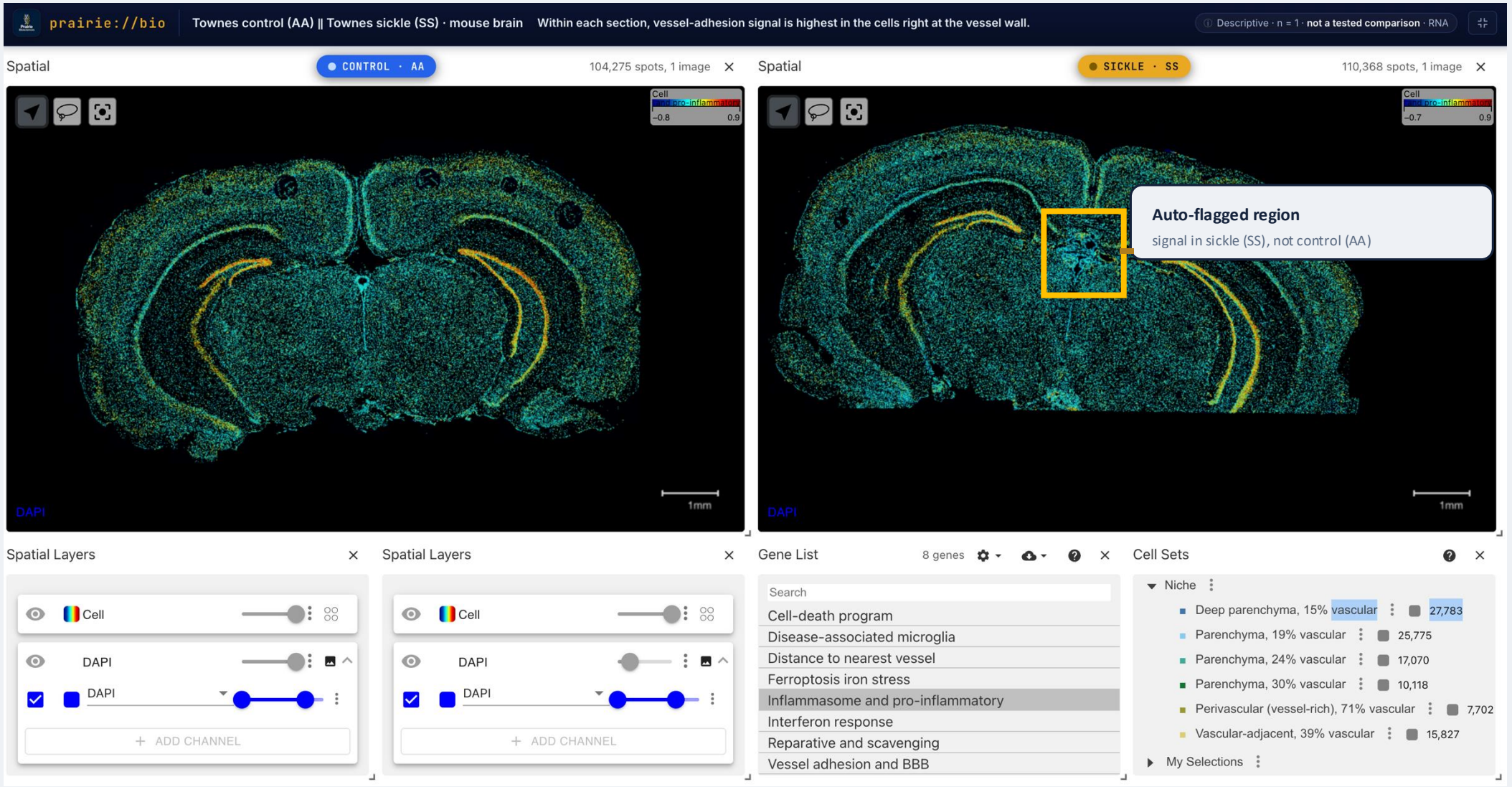


external timestamp on the PRAP → in-toto / Sigstore attestation on every result → open verifier anyone can run.

Compute is already efficient: Prairie RustPipe (open-source) cuts the heavy computing ~17–22×, so AWS credits stretch further. Data separation, controlled compute, HECVAT and HIPAA-aware posture are in place today.

SEE IT RUNNING · 1 OF 2

Live today.



MARKET

A large, underserved market.

Every lab needs an answer it can defend.

TAM

$4.34B

All life-science labs worldwide

avg. $6,007 / project × 630K+ researchers · 14K clinical labs

SAM

$512M

Genomics + tissue mapping · N. America

Analysis across multiple data types

SOM

$20–40M

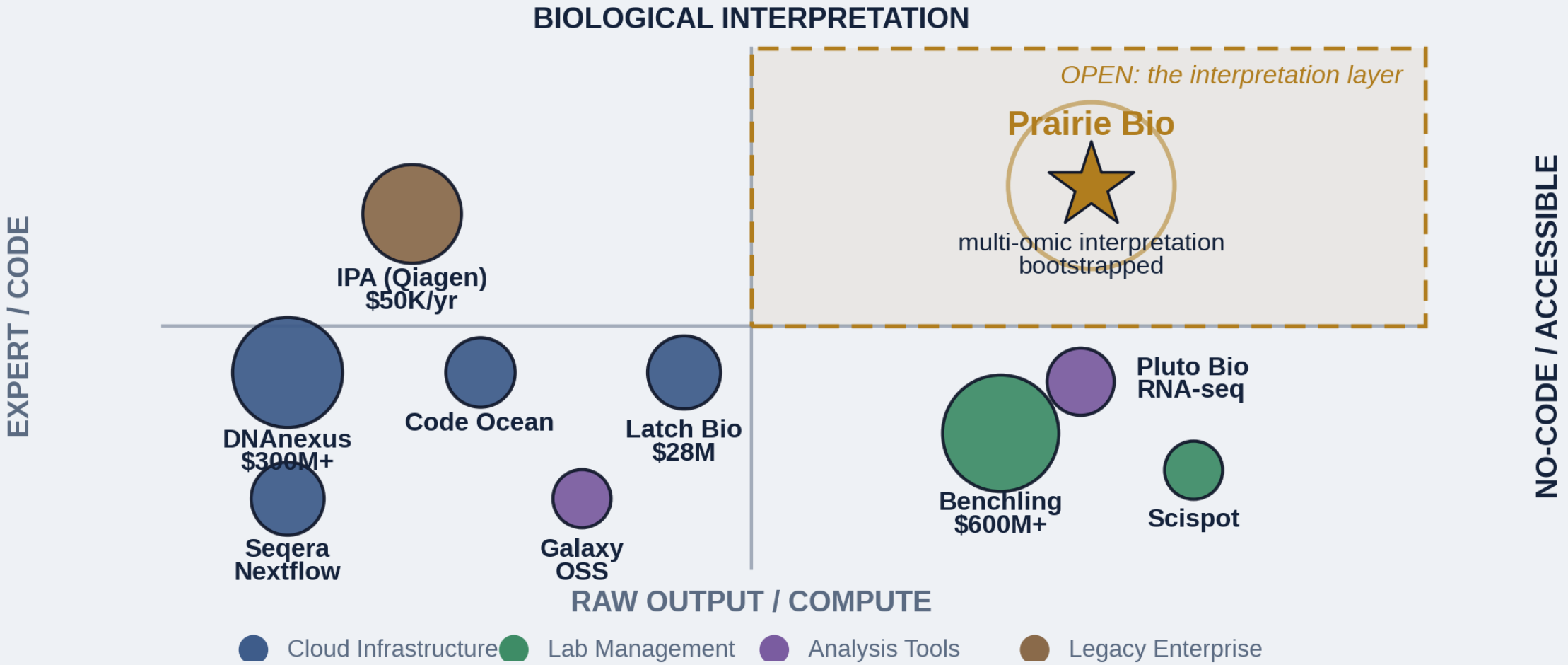
Academic + research-service labs

avg. $6,007 × ~8,000 labs · Year 1–3 reachable

COMPETITION

Where Prairie wins.

$1B+ raised across these tools. None of them check that the analysis fits the experimental design.



REVENUE MODEL

Compliance unlocks customers.

Each compliance step unlocks a bigger buyer — deal size grows ~50×.

NOW	**Academic labs** Research Use Only	**$2K–5K** /analysis
NEAR-TERM	**Biotech & CROs** SOC 2 + HIPAA	**$2K–15K** /project
18 MO	**Pharma & IND** GxP + 21 CFR Part 11	**$25K+** /year
24+ MO	**Pharma / Clinical** Medical-software path if warranted	**$50K+** /enterprise

EVIDENCE

Real science. Real partners. Real IP.

COMPLETED

- ✓ **First-author manuscript, in revision**
  TNBC · with OU HSC (Cherian lab) · targeting npj Precision Oncology
- ✓ **6 academic partnerships**
  OU / OUHSC / OMRF / FSU / UMMC
- ✓ **Multi-omic PDAC engagement delivered**
  UMMC (Mississippi) · CRISPR + RNA-seq + ChIP · under a registered plan
- ✓ **1 Biotech partnership**
  Cadenza Bio

IN PROGRESS

- ○ **Biotech LOI + SOW signed (Syntheia Labs, cardiac organoids) · MSA next**
- ○ **AI drug-discovery collaboration with OU HSC (Stephenson) / Wheeler Bio**
- ○ **SBIR/OCAST proposal submitted**

Platform live on AWS. Every engagement above runs under a Prairie Registered Analysis Plan.

$20K revenue to date · 9 paying projects · 1 first-author manuscript in revision

Prairie RustPipe (open-source) cuts heavy computing ~17-22x, so AWS credits stretch further.

ROADMAP

Land. Expand. Compound.

$150K revenue by end of 2026.

PHASE 1 — Now – EOY 2026

Sell

~$150K revenue

- Sealed PRAP + signed attestation (AWS POC)
- OUHSC: 2 paying → 5 labs
- Grants + fee-per-analysis
- Research platform: SOC2-ready

PHASE 2 — 2027

Sell reliably

~$400K revenue

- Genomics + tissue-mapping pipelines
- Vizgen & STOmics partnerships
- 10 institutional partners
- GxP-validated (21 CFR Part 11)

PHASE 3 — 2028+

Sell at scale

$1M+ revenue

- Vendor channel drives inbound
- Multi-data-type integration moat
- Enterprise / pharma partnerships

Our team.

Founders who span the lab bench and the data; rooted in the Oklahoma research ecosystem.



Joshua Garton, PhD

Founder



Reegan Miller, MS, PhD

Co-Founder

AFFILIATIONS & TRAINING

UT Dallas — The University of Texas at Dallas

Baylor Scott & White Health

KU Medical Center — The University of Kansas

CDC — Centers for Disease Control and Prevention

The University of Oklahoma Health Sciences Center

ARL Bio Pharma

OMRF — Oklahoma Medical Research Foundation

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Appendix.

Technical & financial backup — for the questions diligence asks after the pitch.

A1 Why now: the AI risk A2 Who we serve & channels B Cash & runway

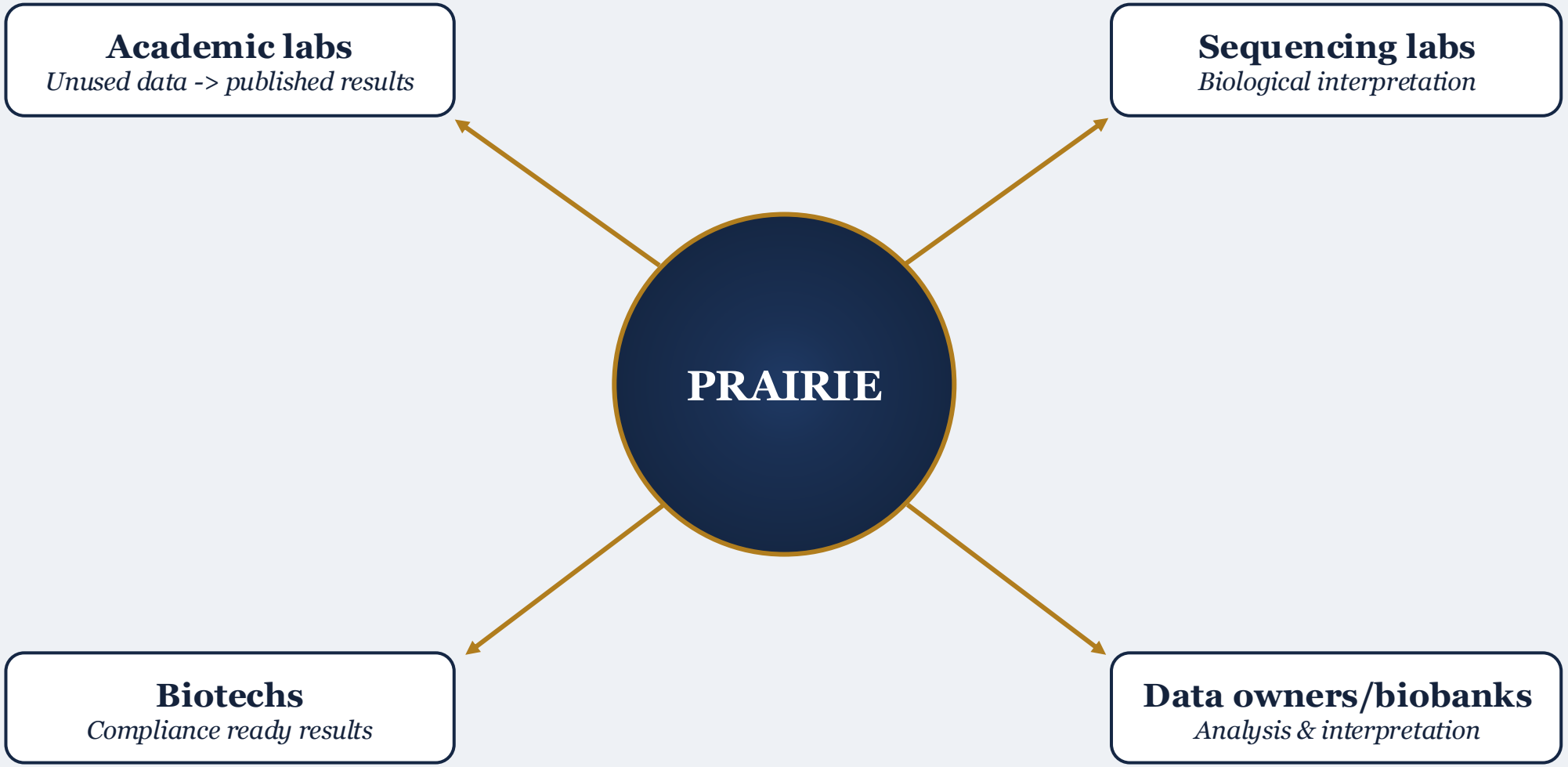
DEMONSTRATED TODAY

Our method, in a first-author TNBC manuscript now in revision.

ON THE ROADMAP

Secure data sharing + a packaged AI-safety product.

Rigorous, trustworthy science that every lab can build on.



GO-TO-MARKET

Three acquisition channels.

Universities & biotechs

Start small, then grow within research universities.

OUHSC · KU Med · OMRF

Research-service labs (CROs)

They generate data but lack analysis — we become their analysis engine.

BioChain — paid pilot in progress

Instrument makers

Vendors send customers who need the data interpreted.

STOmics — inbound to us

HOW IT STAYS TRUSTWORTHY

Why you can trust the answer.

The AI never touches the science — it only turns your question into a fixed, tested recipe.



The AI takes your order — a certified, unchanging process does the science.

APPENDIX A1 · WHY NOW

AI makes a fast wrong answer easy.

Tools that let non-experts run analyses are accelerating the reproducibility crisis.

PROVENANCE TOOLS ALONE

the flight recorder

Faithfully record exactly what happened — including a perfectly traceable wrong answer.

It can't see the batch confound or the mangled genes.

PRAIRIE'S JUDGMENT LAYER

the crash investigator

Catches what the recorder can't: the errors AI-built tools generate at speed.

gene-name corruption · batch · pseudoreplication · default misuse · bad multiple-testing

AI lets scientists generate publishable statistics faster — Prairie keeps them true.

Capital-efficient by design.

Low fixed costs + AWS credits + early revenue = a long runway.

WHAT WE SPEND

Software subscriptions
a few hundred / month

Outreach
~$400, one-time

Hardware (a MacBook)
~$2,500, one-time

Compute (AWS)
$0, $10K credits last into 2027

Prairie RustPipe (open-source) cuts the heavy computing ~17-22x, so AWS credits stretch further.

Operating burn: low hundreds per month.

Illustrative; exact figures being finalized from receipts.

THE 2026 PICTURE

Revenue target	~$150K
Cash operating costs	~$5K

WHERE THE $25K AWS POC GOES

1. Sealed PRAP: external timestamp before data access
2. Signed attestations (in-toto / Sigstore)
3. Open-source verifier, no Prairie account needed
4. First agent-run result with a checkable certificate

The POC funds the layer that lets someone who doesn't trust us verify the result anyway.